

Some Ramblings on Set Theory

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Abstract

This paper aims to serve as an expository article to provide the reader with a good understanding of ZFC Set Theory, the varied applications of the Axiom of Choice and the various statements that it is equivalent to. Basically, I just want to show the reader how cool set theory really can be.

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§1 Introduction

This section may be skipped if the reader is already acquainted with basic set theoretic definitions and notations.

§1.1 Partially Ordered Sets

We write $x \leq y$ when (x, y) belongs to the relation. The pair $(P, \leq)$ is a *partially ordered set*, or *poset*, if for all $x, y, z \in P$:

1. $x \leq x$ (reflexivity);
2. $x \leq y$ and $y \leq x$ imply $x = y$ (antisymmetry);
3. $x \leq y$ and $y \leq z$ imply $x \leq z$ (transitivity).

We write $x < y$ when $x \leq y$ and $x \neq y$. Two elements are *comparable* if $x \leq y$ or $y \leq x$. A partial order is a *total order* when every two elements are comparable. Every subset $S \subseteq P$ carries the order obtained by restricting $\leq$ to S .

A subset $C \subseteq P$ is a *chain* if every two elements of C are comparable. The empty set and every one-element subset are chains. A chain C is a *maximal chain* if there is no chain D with $C \subsetneq D \subseteq P$.

Let S be a subset of a poset $(P, \leq)$. An element $u \in P$ is an *upper bound* of S if

$$s \leq u \quad \text{for every } s \in S.$$

The upper bound need not belong to S . If $s < u$ for every $s \in S$, then u is called a *strict upper bound*. A *lower bound* is defined similarly by requiring $u \leq s$ for every $s \in S$.

If an upper bound belongs to S , then it is the greatest element of S . Thus “upper bound” describes an element of the surrounding poset P , while “greatest element” must be a member of S itself.

An element $g \in S$ is the *greatest element* of S if $s \leq g$ for every $s \in S$. An element $m \in S$ is *maximal in* S if no $s \in S$ satisfies $m < s$. Least and minimal elements are defined by reversing the inequalities.

A greatest element is always maximal and, if it exists, is unique. A maximal element need not be comparable with every other element, so a poset may have several maximal elements and no greatest element. Likewise, an upper bound of S need not belong to S , whereas a greatest element must belong to S .

The power set $\mathcal{P}(X)$ is a poset under inclusion. Its chains are precisely the families of subsets nested by inclusion, and X is its greatest element. By contrast, in the poset

$$P = \{\{0\}, \{1\}\}$$

under inclusion, the two elements are incomparable: $\{0\} \not\subseteq \{1\}$ and $\{1\} \not\subseteq \{0\}$. Nevertheless, each is maximal. Indeed, there is no member of P that properly contains $\{0\}$, and there is no member of P that properly contains $\{1\}$.

There is no greatest element, however. A greatest element would have to contain every member of P ; in particular, it would have to contain both $\{0\}$ and $\{1\}$. Neither member of

P does so. The set $\{0, 1\}$ would be an upper bound in the larger poset $\mathcal{P}(\{0, 1\})$, but it is not an element of P and therefore cannot be the greatest element of P .

§1.2 Functions and Fibres

Let $f : X \rightarrow Y$ be a function and let $y \in Y$. The *fibre of f over y* is the set of all elements of the domain that are sent to y :

$$f^{-1}(\{y\}) = \{x \in X : f(x) = y\}.$$

A fibre may be empty. The function f is surjective precisely when every fibre over an element of Y is nonempty. The nonempty fibres of a function are pairwise disjoint and together cover X , so they form a partition of the domain.

Define $f : \mathbb{Z} \rightarrow \{0, 1\}$ by letting $f(n)$ be the remainder when n is divided by 2. The fibre over 0 is the set of even integers, while the fibre over 1 is the set of odd integers. These two fibres partition $\mathbb{Z}$.

§1.3 Tagged Sets

Let $\{X_i : i \in I\}$ be a family of sets. The *tagged copy* of X_i is

$$\{i\} \times X_i = \{(i, x) : x \in X_i\}.$$

The first coordinate i is called the *tag*; it records which set the element came from. Tagged copies belonging to different indices are disjoint, even when the original sets overlap.

Suppose $X_0 = \{a, b\}$ and $X_1 = \{b, c\}$. The element b belongs to both original sets, but the tagged elements $(0, b)$ and $(1, b)$ are different. Thus the tagged copies

$$\{0\} \times X_0 = \{(0, a), (0, b)\}, \quad \{1\} \times X_1 = \{(1, b), (1, c)\}$$

are disjoint.

§1.4 Partitions and Transversals

Let X be a set. A *partition* of X is a collection $\mathcal{P}$ of nonempty subsets of X such that every element of X belongs to exactly one member of $\mathcal{P}$. The members of $\mathcal{P}$ are called the *blocks* of the partition. Equivalently, distinct blocks are disjoint and their union is X .

A *transversal*, sometimes called a *choice set*, for the partition $\mathcal{P}$ is a subset $T \subseteq X$ containing exactly one element from each block. In symbols,

$$|T \cap B| = 1 \quad \text{for every } B \in \mathcal{P}.$$

The relevant term is *transversal*, rather than “traversal”, which actually intuitively makes more sense because if you think about it, you’re essentially traversing through the partitions of a set, but instead some evil people decided to name it “transversal”.

The collection

$$\mathcal{P} = \{\{1, 2\}, \{3, 4\}, \{5\}\}$$

is a partition of $\{1, 2, 3, 4, 5\}$. The set $\{1, 4, 5\}$ is a transversal because it contains one element from each of the three blocks. The set $\{1, 2, 5\}$ is not a transversal: it contains two elements from the first block and none from the second.

Ordinals extend the natural numbers so that positions in infinite well-ordered lists can also be named.

Well-orders and order types.

A total order is a *well-order* if every nonempty subset $S \subseteq X$ has a least element: some $s \in S$ such that $s \leq t$ for every $t \in S$.

For example, $\mathbb{N}$ with its usual order is well-ordered.

§1.5 Order isomorphism

An *order isomorphism* between ordered sets $(X, \leq_X)$ and $(Y, \leq_Y)$ is a bijection $f : X \rightarrow Y$ such that, for all $x, x' \in X$,

$$x \leq_X x' \iff f(x) \leq_Y f(x').$$

Thus f both preserves and reflects the order. Two well-ordered sets have the same *order type* precisely when an order isomorphism exists between them.

For example,

$$a < b < c \quad \text{and} \quad 10 < 20 < 30$$

have the same order type.

An ordinal is the standard set used to represent one such order type.

Theorem (Classification of Well-Orderings). Let $(W, \leq)$ be a well-ordered set. Then there exists a unique ordinal α such that $(W, \leq) \cong (\alpha, \in)$.

§1.5.1 Preliminary Lemmas

Before proving the main theorem, we recall two standard properties of well-ordered sets and ordinals.

Strict Increase of Order Embeddings: If $(X, \leq_X)$ is a well-ordered set and $g : X \rightarrow X$ is an order embedding (i.e., $x_1 \leq_X x_2 \iff g(x_1) \leq_X g(x_2)$), then $x \leq_X g(x)$ for all $x \in X$.

Proof. Suppose the set $A = \{x \in X : g(x) <_X x\}$ is non-empty. Since X is well-ordered, A has a least element $x_0 \in A$. Since $g(x_0) <_X x_0$ and g is order-preserving, applying g yields $g(g(x_0)) <_X g(x_0)$. This implies $g(x_0) \in A$. However, $g(x_0) <_X x_0$, which contradicts x_0 being the minimum of A . Hence, $A = \emptyset$. $\square$

Initial Segments of Ordinals: An initial segment of an ordinal α generated by an element $\beta \in \alpha$ is equal to β itself. That is, $\text{pred}(\alpha, \beta) = \{\gamma \in \alpha : \gamma \in \beta\} = \beta$.

§1.5.2 Proof of Uniqueness

Theorem (Uniqueness). Let W be a well-ordered set, and let α and β be ordinals. If $W \cong \alpha$ and $W \cong \beta$, then $\alpha = \beta$.

Proof. By the transitivity of order-isomorphisms, $W \cong \alpha$ and $W \cong \beta$ together imply that there exists an order-isomorphism $f : \alpha \rightarrow \beta$.

Consider f as an order-embedding from α into β :

- For any $\gamma \in \alpha$, since $f(\gamma) \in \beta$, Lemma 1.5.1 implies $\gamma \leq f(\gamma)$, which means $\gamma \subseteq f(\gamma) \in \beta$. Thus $\gamma \in \beta$, so $\alpha \subseteq \beta$.
- Applying the same logic to the inverse isomorphism $f^{-1} : \beta \rightarrow \alpha$, we obtain $\beta \subseteq \alpha$.

From $\alpha \subseteq \beta$ and $\beta \subseteq \alpha$, we conclude that $\alpha = \beta$. □

§1.5.3 Proof of Existence

Let $(W, \leq)$ be a well-ordered set. For each $x \in W$, define the initial segment generated by x as:

$$W_x = \{y \in W : y < x\}.$$

Define the set $S \subseteq W$ of elements whose initial segments are isomorphic to some ordinal:

$$S = \{x \in W : \exists \text{ ordinal } \alpha_x \text{ such that } W_x \cong \alpha_x\}.$$

By uniqueness, for each $x \in S$, the ordinal α_x is unique. Thus, we have a well-defined mapping $f : S \rightarrow \text{On}$ given by $f(x) = \alpha_x$.

Lemma. S is an initial segment of W .

Proof. Let $x \in S$ and let $y < x$. Since $x \in S$, there is an isomorphism $g : W_x \rightarrow \alpha_x$. The restriction of g to W_y gives an isomorphism between W_y and an initial segment of α_x . By Lemma 1.5.1, any initial segment of α_x is an ordinal $\beta \in \alpha_x$. Thus $W_y \cong \beta$, which proves $y \in S$. □

Lemma. *Claim:* $S = W$.

Proof. Assume for contradiction that $S \neq W$. Since W is well-ordered, $W \setminus S$ contains a least element $m = \min(W \setminus S)$.

Since m is the minimal element outside S , every element strictly smaller than m is in S . Therefore, $W_m = S$.

For each $y \in W_m$, $f(y) = \alpha_y$. The map $f : W_m \rightarrow \text{On}$ defined by $y \mapsto \alpha_y$ is an order-isomorphism onto its image:

$$\alpha = \{f(y) : y < m\} = \{\alpha_y : y < m\}.$$

Since α is a transitive set of ordinals, α is itself an ordinal. Consequently, $W_m \cong \alpha$, which implies $m \in S$. This directly contradicts $m \in W \setminus S$.

Thus, $S = W$. □

Since $S = W$, every initial segment W_x is isomorphic to an ordinal $f(x) = \alpha_x$. The function $f : W \rightarrow \text{On}$ defined by $f(x) = \alpha_x$ forms an order-isomorphism from W onto its image:

$$\gamma = \{f(x) : x \in W\}.$$

Because γ is a transitive set of ordinals, γ is an ordinal. Hence, $W \cong \gamma$.

This establishes that W is order-isomorphic to the ordinal γ , completing the proof.

§2 Ordinals

Ordinals as sets. In the von Neumann representation, each ordinal is the set of all smaller ordinals.

The first few are

$$0 = \emptyset, \quad 1 = \{0\}, \quad 2 = \{0, 1\}, \quad 3 = \{0, 1, 2\}.$$

A set T is *transitive* when $x \in y \in T$ implies $x \in T$.

In general, an ordinal is a transitive set that is well-ordered by membership. For ordinals, membership records order:

$$\alpha < \beta \iff \alpha \in \beta.$$

Consequently, the elements of an ordinal β are exactly the ordinals that occur before β .

The first infinite ordinal is

$$\omega = \{0, 1, 2, 3, \dots\},$$

the order type of the natural numbers. If one new position is placed after all natural-number positions, the result is $\omega + 1$.

$$\omega + 1 = \{0, 1, 2, 3, \dots, \omega\}$$

Although ω and $\omega + 1$ have the same cardinality, they are different ordinals: the latter has a greatest element and the former does not. This illustrates the distinction between *cardinality*, which measures size, and an ordinal, which records a well-ordering. Thus we can differentiate between unequal sets that have the same cardinality but different "size" in some sense.

Successor and limit ordinals.

The ordinal immediately after α is its *successor*

$$\alpha + 1 = \alpha \cup \{\alpha\}.$$

A nonzero ordinal that is not a successor is a *limit ordinal*. A limit ordinal contains all earlier stages and satisfies

$$\lambda = \bigcup_{\beta < \lambda} \beta.$$

Every ordinal is therefore either 0, a successor ordinal, or a nonzero limit ordinal.

For example, ω is a limit ordinal because no largest finite ordinal occurs immediately before it.

Any set A of ordinals has a least upper bound,

$$\sup A = \bigcup A.$$

If an ordinal strictly larger than every member of A is required, one may instead use $\sup\{\alpha + 1 : \alpha \in A\}$. This construction is used in Hartogs' lemma below.

Transfinite induction and recursion.

To prove a statement $P(\alpha)$ for every ordinal, it is enough to show that, whenever $P(\beta)$ holds for all $\beta < \alpha$, it also holds for α . This is *transfinite induction*.

Similarly, *transfinite recursion* defines the value at stage α from the values at all stages $\beta < \alpha$. At a successor stage this includes the immediately preceding value; at a limit stage there is no immediate predecessor, so all earlier values are used.

These principles work because every nonempty collection of ordinals has a least member, and they will be used repeatedly below.

Two sets have the same *cardinality*, written $|X| = |Y|$, if a bijection $X \rightarrow Y$ exists. We write $|X| \leq |Y|$ if an injection $X \rightarrow Y$ exists, and $|X| < |Y|$ if $|X| \leq |Y|$ but $|X| \neq |Y|$. A set is *finite* if it is in bijection with $n = \{0, 1, \dots, n-1\}$ for some natural number n ; otherwise it is *infinite*.

An ordinal κ is an *initial ordinal*, or cardinal number, if no smaller ordinal is in bijection with it. Every well-orderable set is in bijection with a unique initial ordinal, which represents its cardinality. The first infinite initial ordinal is denoted $\aleph_0$ and is represented by ω .

§2.1 Hartogs' Lemma

Hartogs' lemma:

For every set X , there is an ordinal θ for which no injection $\theta \rightarrow X$ exists.

Proof. Consider all well-orderings of subsets of X . They form a set, since their underlying sets and relations are subsets of X and $X \times X$, respectively. The standard theorem that every well-ordered set has a unique ordinal order type gives a set H of all these order types. Let

$$\theta = \sup\{\alpha + 1 : \alpha \in H\}.$$

Here $\alpha + 1$ is the next ordinal after α , and the supremum is the least ordinal not smaller than any member of the displayed set; it exists as the union of that set of ordinals. Thus $\theta > \alpha$ for every $\alpha \in H$. An injection $\theta \rightarrow X$ would transfer the well-order of θ to a subset of X , putting θ in H , a contradiction. This proves the lemma. ■

§3 The Axioms of ZF Set Theory

Zermelo-Fraenkel set theory, abbreviated *ZF*, is the current most widely accepted model of set theory that most people work in. We shall be using this model for this article, so I decided it would be a good idea to give my readers an idea of the axioms of ZF.

The axioms of ZF:

1. **Extensionality.** If there exist sets A and B such that

$$x \in A \iff x \in B$$

Then we say $A = B$

2. **Empty Set.** There is a set having no elements: It is denoted by $\emptyset$.
3. **Pairing.** Given sets a and b , the set $\{a, b\}$ exists:

4. **Union.** For every set A , the set containing all elements of members of A exists. This set is denoted by $\bigcup A$.
5. **Power Set.** For every set A , the set of all subsets of A exists. This set is denoted by $\mathcal{P}(A)$.
6. **Infinity.** There is a set I containing $\emptyset$ and containing $x \cup \{x\}$ whenever it contains x . This provides an infinite set and allows the natural numbers to be constructed.
7. **Separation Schema.** If A is a set and $\varphi(x)$ is any property expressible in the language of set theory, then

$$\{x \in A : \varphi(x)\}$$

is a set.

8. **Replacement Schema.** If a formula $\varphi(x, y)$ assigns a unique set y to every $x \in A$, then the collection of all assigned values is a set.
9. **Foundation.** Every nonempty set has a member disjoint from it:

$$\forall A \text{ such that } A \neq \emptyset, \exists x \in A \text{ such that } x \cap A = \emptyset.$$

ZF does not include the Axiom of Choice. The theory obtained by adding AC to ZF is called *ZFC*.

§4 The Cantor–Schröder–Bernstein Theorem

Cantor–Schröder–Bernstein theorem:

Let A and B be sets. If there is an injection $f : A \rightarrow B$ and an injection $g : B \rightarrow A$, then there is a bijection $h : A \rightarrow B$. Equivalently,

$$|A| \leq |B| \quad \text{and} \quad |B| \leq |A| \quad \implies \quad |A| = |B|.$$

This theorem is provable in ZF alone. It should not be confused with *cardinal comparability*, which says that for any two sets there is an injection in at least one direction and is equivalent to AC.

Proof. Let $f : A \rightarrow B$ and $g : B \rightarrow A$ be injections. Write

$$g[B] = \{g(b) : b \in B\}$$

for the image of B under g . We first identify the elements of A that will be sent forward using f . Define

$$A_0 = A \setminus g[B], \quad A_{n+1} = g[f[A_n]] \quad (n \in \mathbb{N}),$$

and let

$$C = \bigcup_{n \in \mathbb{N}} A_n.$$

Thus A_0 consists of the elements of A that are not reached by g . Each later A_{n+1} is obtained by moving the preceding layer across to B using f and then back to A using g .

Define $h : A \rightarrow B$ by

$$h(a) = \begin{cases} f(a), & a \in C, \\ g^{-1}(a), & a \notin C. \end{cases}$$

This is well-defined. If $a \notin C$, then in particular $a \notin A_0 = A \setminus g[B]$, so $a \in g[B]$. Because g is injective, there is a unique $b \in B$ with $g(b) = a$; this unique element is what $g^{-1}(a)$ denotes.

We show first that h is injective. On C it agrees with the injection f , and on $A \setminus C$ it agrees with the inverse of the injection g . It remains only to rule out a collision between the two pieces. Suppose $a \in C$, $a' \notin C$, and $f(a) = g^{-1}(a')$. Applying g gives

$$a' = g(f(a)).$$

Since $a \in A_n$ for some n , the right-hand side belongs to $A_{n+1} \subseteq C$, contradicting $a' \notin C$. Hence h is injective.

Finally, let $b \in B$. If $g(b) \notin C$, then

$$h(g(b)) = g^{-1}(g(b)) = b.$$

If $g(b) \in C$, it cannot belong to A_0 , because it lies in the image of g . Therefore $g(b) \in A_{n+1}$ for some n . By the definition of A_{n+1} , there is an $a \in A_n$ such that

$$g(b) = g(f(a)).$$

The injectivity of g gives $b = f(a)$, and since $a \in A_n \subseteq C$, we have $h(a) = f(a) = b$. Thus every element of B has a preimage, so h is surjective. Therefore h is a bijection.

§5 Different Forms of the Axiom of Choice

The Axiom of Choice can be written in several forms that look different but express the same ability to make a simultaneous family of choices.

Direct formulations of the Axiom of Choice:

Over ZF, the following statements are equivalent:

1. **Product form (AC):** For every family $\{X_i : i \in I\}$ of nonempty sets,

$$\prod_{i \in I} X_i \neq \emptyset.$$

2. **Family choice-function form:** For every family $\{X_i : i \in I\}$ of nonempty sets, there is a function f on I such that $f(i) \in X_i$ for every $i \in I$.

3. **Power-set choice-function form (AC'):** For every nonempty set X , there is a function

$$c : \mathcal{P}(X) \setminus \{\emptyset\} \longrightarrow X$$

such that $c(A) \in A$ for every nonempty subset $A \subseteq X$.

4. **Right-inverse form:** Every surjective function $p : E \rightarrow I$ has a right inverse; that is, there is a function $s : I \rightarrow E$ such that $p \circ s = \text{id}_I$.

5. **Transversal form:** Every partition has a transversal.

Equivalence of the formulations. The first two statements merely describe the same object in different notation: an element of $\prod_{i \in I} X_i$ is precisely a function f satisfying $f(i) \in X_i$ for every i .

The family form gives AC' by taking the family of all nonempty subsets of a fixed set X . Conversely, given AC' and a family $\{X_i : i \in I\}$, the empty function handles the case $I = \emptyset$. Otherwise apply AC' to $X = \bigcup_{i \in I} X_i$ and define $f(i) = c(X_i)$. Thus the first three forms

are equivalent.

Suppose the family form holds and $p : E \rightarrow I$ is surjective. Every fibre $p^{-1}(\{i\})$ is nonempty, so choosing one point $s(i)$ from each fibre produces a right inverse of p . Conversely, for a family $\{X_i : i \in I\}$, form its disjoint union

$$E = \{(i, x) : i \in I \text{ and } x \in X_i\}.$$

The projection $p(i, x) = i$ is surjective. A right inverse has the form $s(i) = (i, f(i))$ and therefore supplies the desired choice function.

Finally, the blocks of a partition form a family of pairwise disjoint nonempty sets, and a choice function for those blocks is exactly a transversal. In the other direction, the tagged sets $\{i\} \times X_i$ partition the disjoint union displayed above; a transversal chooses one tagged element from each X_i . Hence all five statements are equivalent.

§5.1 Weaker Choice Principles

Full AC permits choices from an arbitrary family of arbitrary nonempty sets. It can be weakened by restricting the size of the indexing set, restricting the sets from which choices are made, or requiring choices only along one step-by-step construction.

Countable Choice (AC_ω). For every sequence $\{X_n : n \in \mathbb{N}\}$ of nonempty sets, there is a function f such that $f(n) \in X_n$ for every $n \in \mathbb{N}$. Equivalently,

$$\prod_{n \in \mathbb{N}} X_n \neq \emptyset.$$

This is AC restricted to families indexed by the natural numbers.

Countable Choice for finite sets ($\text{AC}_\omega^{\text{fin}}$). For every sequence $\{A_n : n \in \mathbb{N}\}$ of nonempty finite sets, there is a function f such that $f(n) \in A_n$ for every $n \in \mathbb{N}$. This restricts both the index set and the members of the family.

A *rooted tree* consists of vertices arranged in levels, beginning with a root, so that every non-root vertex has a unique immediate predecessor. It is *finitely branching* if every vertex has only finitely many immediate successors. An *infinite branch* passes through one vertex at every level, moving to an immediate successor at each step.

König's infinity lemma:

Over ZF, the following statements are equivalent:

1. Countable Choice for finite sets $\text{AC}_\omega^{\text{fin}}$.
2. Every infinite finitely branching rooted tree has an infinite branch.

Why these two statements are equivalent. Assume $\text{AC}_\omega^{\text{fin}}$ and let T be an infinite finitely branching rooted tree. Each level L_n is finite and nonempty. The set of linear orders on L_n is also finite and nonempty, so countable choice for finite sets lets us fix one such order on every level.

Call a vertex *live* if it has descendants at arbitrarily high levels. The root is live, and a live

vertex has a live immediate successor: otherwise its finitely many successors would all stop after finitely many levels. Starting at the root, choose at each stage the least live successor in the fixed order on the next level. Recursion then produces an infinite branch.

Conversely, let $\{A_n : n \in \mathbb{N}\}$ be nonempty finite sets. Make a tree whose level n consists of all partial choices

$$(a_0, \dots, a_{n-1}) \in \prod_{k < n} A_k.$$

Every level is finite and nonempty, and each vertex has only finitely many extensions to the next level. An infinite branch gives a sequence $(a_n)_{n \in \mathbb{N}}$ with $a_n \in A_n$, which is the required choice function.

Dependent Choice (DC). Let X be nonempty and let R be a binary relation on X such that for every $x \in X$ there is some $y \in X$ with xRy . Then, for every starting point $x_0 \in X$, there is a sequence $(x_n)_{n \in \mathbb{N}}$ satisfying

$$x_n R x_{n+1} \quad \text{for every } n \in \mathbb{N}.$$

Unlike ordinary countable choice, the available choice at one stage may depend on the choice made at the preceding stage.

The elementary implication chain is

$$\text{AC} \implies \text{DC} \implies \text{AC}_\omega \implies \text{AC}_\omega^{\text{fin}}.$$

For example, AC proves DC by choosing one permitted successor for each $x \in X$ and then iterating that choice. DC proves countable choice by applying it to finite partial choice functions, ordered by one-step extension. The remaining implication is immediate by restriction to finite sets.

None of the reverse implications can be proved in ZF. Thus these statements are genuine weakenings of AC and should not be placed in the main list of principles equivalent to full choice.

§6 Some Major Equivalent Principles

The following are equivalent formulations of the Axiom of Choice.

Equivalent forms of the Axiom of Choice:

1. **Cardinal comparability:** For any sets A and B , there is an injection $A \rightarrow B$ or an injection $B \rightarrow A$.
2. **Tarski's theorem:** For every infinite set A , there is a bijection

$$A \longrightarrow A \times A.$$

3. **Well-ordering theorem:** Every set can be well-ordered.
4. **Zorn's lemma:** Every nonempty partially ordered set in which every chain has an upper bound has a maximal element.
5. **Hausdorff's maximal principle:** Every partially ordered set has a maximal chain.

6. **Chain-extension form:** Every chain in a partially ordered set is contained in a maximal chain.

§7 Applications of AC outside of Set Theory

§7.1 Every Vector Space has a Basis

The vector-space basis theorem:

Let X be any vector space over any field F , and assume Zorn's lemma. Then if L is a linearly independent subset of X , there exists a basis of X containing L . In particular, X does have a basis at all.

Proof. Let $\mathcal{A}$ be the set of linearly independent subsets of X containing L (in particular, $\mathcal{A}$ is not empty), then $\mathcal{A}$ is partially ordered by inclusion. For each chain $\mathcal{C} \subseteq \mathcal{A}$, define $\hat{\mathcal{C}} = \bigcup \mathcal{C}$. Clearly, $\hat{\mathcal{C}}$ is an upper bound of $\mathcal{C}$. Next we show that $\hat{\mathcal{C}} \in \mathcal{A}$. Let $V := \{v_1, \dots, v_n\} \subseteq \hat{\mathcal{C}}$ be a finite collection of vectors. Then there exist sets $C_1, \dots, C_n \in \mathcal{C}$ such that $v_i \in C_i$ for all $1 \leq i \leq n$. Since $\mathcal{C}$ is a chain, there is a number k with $1 \leq k \leq n$ such that

$$C_k = \bigcup_{i=1}^n C_i$$

and thus $V \subseteq C_k$, that is, V is linearly independent. Therefore, $\hat{\mathcal{C}}$ is an element of $\mathcal{A}$. According to Zorn's lemma, $\mathcal{A}$ has a maximal element M , which is linearly independent. We show now that M is a basis. Let $\langle M \rangle$ be the span of M . Assume there exists an $x \in X \setminus \langle M \rangle$. Let $\{x_1, \dots, x_n\} \subseteq M$ be a finite collection of vectors and $a_1, \dots, a_{n+1} \in F$ elements such that

$$a_1x_1 + \dots + a_nx_n - a_{n+1}x = 0.$$

If a_{n+1} was necessarily zero, so would be the other a_i , $1 \leq i \leq n$, making $\{x\} \cup M$ linearly independent, in contradiction to the maximality of M . If $a_{n+1} \neq 0$, we would have

$$x = \frac{a_1}{a_{n+1}}x_1 + \dots + \frac{a_n}{a_{n+1}}x_n,$$

contradicting $x \notin \langle M \rangle$. Thus such an x does not exist and $X = \langle M \rangle$, so M is a generating set and hence a basis.

Taking $L = \emptyset$, we see that X does have a basis at all. $\square$

§7.2 Some other interesting results/ equivalents of some form of AC

Results:

1. The Axiom of Choice.
2. Every nonempty set admits a group structure.
3. Every connected graph has a spanning tree; equivalently, every tree in a connected graph can be extended to a spanning tree, while every spanning subgraph contains a spanning tree
4. Banach-Tarski Paradox: a solid unit ball in $\mathbb{R}^3$ can be decomposed into two solid balls.